



2025

T24

BOOKLET NO.

112180

Forest Services

Chemistry

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

- Q1. Write short notes on **any five** of the following. (5×8=40)
- Q1. (a) Classification of Ligands and Effective Atomic Number (EAN) rule. 8
- Q1. (b) First and Second Laws of Thermodynamics. 8
- Q1. (c) Fuel cells and batteries. 8
- Q1. (d) Collision Theory and Transition State Theory. 8
- Q1. (e) Quantum numbers and shapes of s, p and d orbitals. 8
- Q1. (f) Excess thermodynamic functions and their determination. 8
- Q1. (g) Born-Haber cycle and Valence Bond Theory. 8
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- Q2. (a) Explain isomerism in coordination compounds. Describe Crystal Field Theory, crystal field splitting in octahedral complexes and its modifications. Explain trans effect and its theory with suitable examples. 15
- Q2. (b) State and explain Third Law of Thermodynamics. Discuss free energy function and entropy as a state function. Derive the relation between Gibbs free energy and equilibrium constant. 15
- Q2. (c) Derive the Clausius-Clapeyron equation and explain its significance in phase equilibrium. The vapour pressure of a liquid is 200 mm Hg at 300 K and 400 mm Hg at 320 K. Calculate the enthalpy of vaporization. 10
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- Q3. (a) Explain the principle, instrumentation and applications of Polarography.
The standard electrode potentials are :
 $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$
 $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$
Calculate the standard EMF of the Daniell cell. 15
- Q3. (b) Derive the differential and integrated rate equations for zero and first order reactions. Explain the characteristics of each order reaction. For a first order reaction, rate constant $k = 2.3 \times 10^{-3} \text{ s}^{-1}$. Calculate the half-life of the reaction. 15
- Q3. (c) Write the time-independent Schrödinger wave equation and explain its physical significance. Derive the expression for energy of a particle in a one-dimensional box. 10
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- Q4. (a) Describe Molecular Orbital Theory for diatomic molecules. The dipole moment of HCl is 1.08 D and bond length is 1.27 Å. Calculate the percentage ionic character. 15
- Q4. (b) Discuss lanthanide contraction and magnetic and spectral properties of lanthanides. Write the important applications of lanthanides. Calculate the magnetic moment of Gd^{3+} ion. 15
- Q4. (c) Explain Langmuir adsorption isotherm and BET adsorption isotherm. X-rays of wavelength 1.54 Å are diffracted by a crystal at an angle of 30° . Calculate the interplanar spacing using Bragg's law. 10
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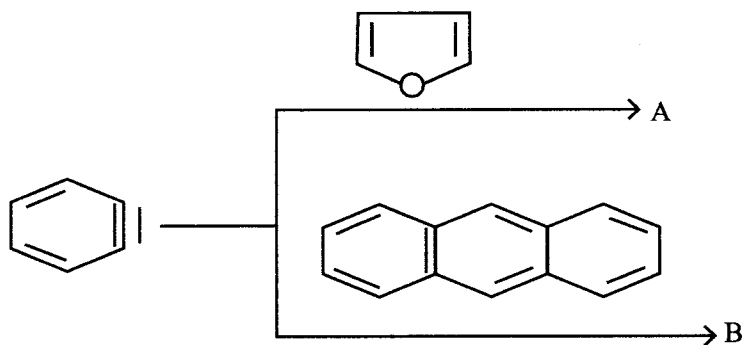
SECTION – B

Q5. Solve any five out of seven.

(5×8=40)

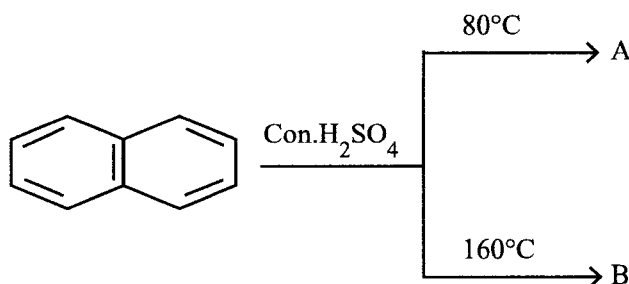
- Q5. (a) i) What is trapping of an intermediate ? Write A and B products formed in the following reaction.

4



- ii) Write A and B products formed in the following reaction and explain the sulphonation of naphthalene at different temperature.

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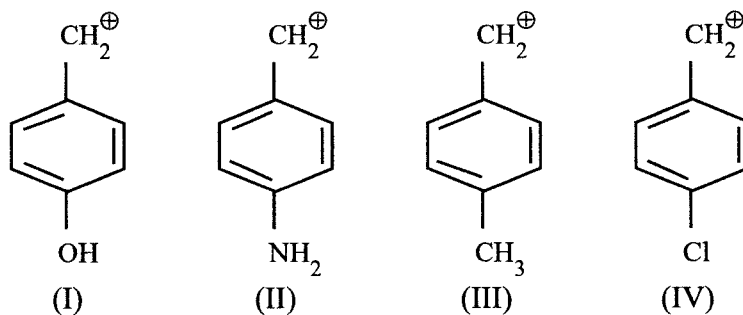


- Q5. (b) i) Explain the factors affecting the stability of carbanion.

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- ii) Arrange the following intermediates in decreasing order of stability and justify your answer.

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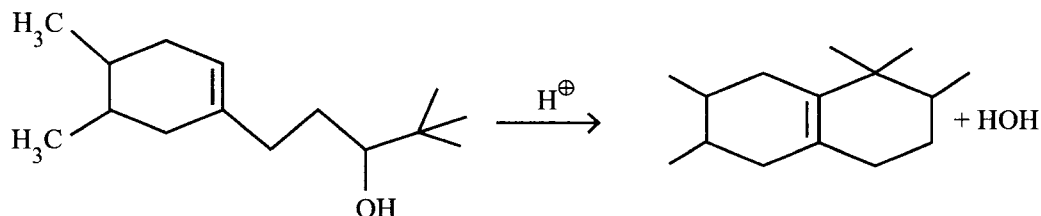




Q5. (c) i) Cis isomer of 4-tertiary butyl cyclohexyl tosylate undergoes faster E_2 while trans isomer undergoes slower E_1 . Explain. 4

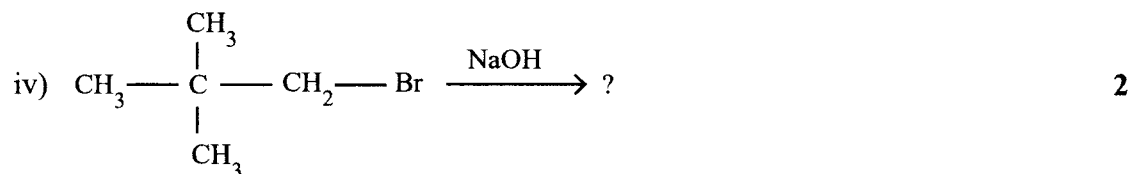
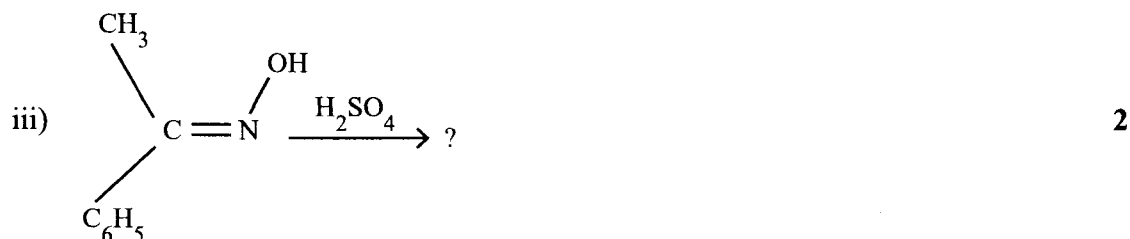
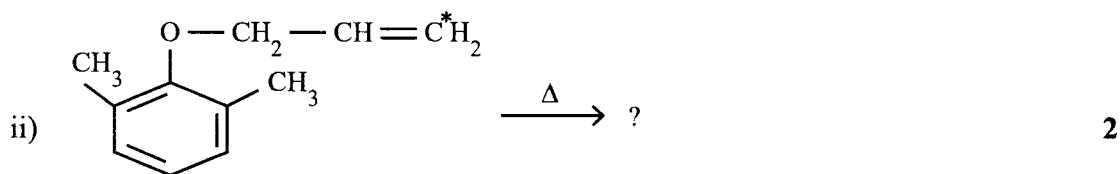
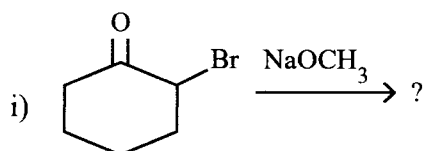
ii) Explain E_1 CB mechanism with suitable example. 4

Q5. (d) i) Give the mechanism of the given reaction. 4



ii) Explain the aromaticity of sydnones. 4

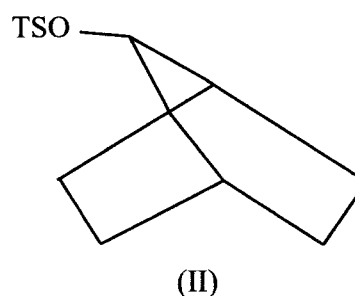
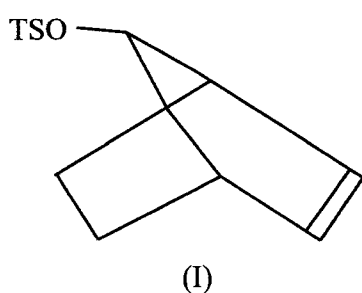
Q5. (e) Name the following rearrangement reactions and write the major product formed in each case. (2×4=8)





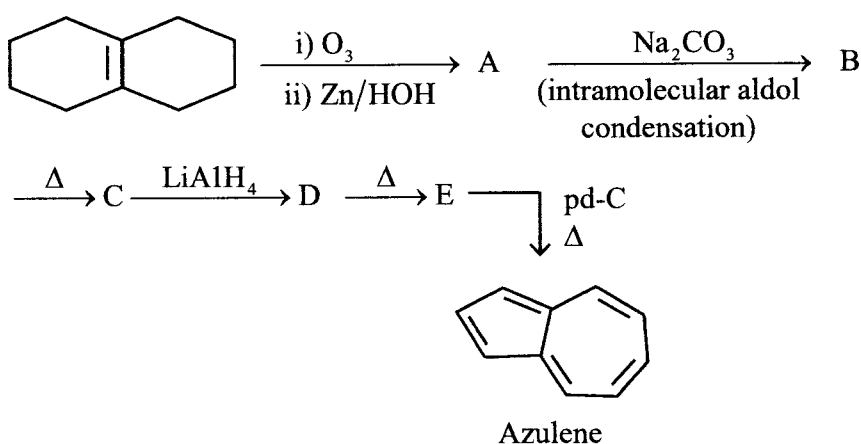
- Q5. (f) i) What is Reformatsky reaction ? Give its mechanism. 4
ii) What is ESR spectroscopy ? Explain its basic principle. 4

- Q5. (g) i) Acetolysis of 7-norbornenyl tosylate (I) is 10^{11} times faster than that of the saturated analogue 7-norbornenyl tosylate (II). Explain. 4

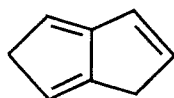


- ii) What is sedimentation method for the determination of molecular weight of polymers. 4

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- Q6. (a) i) Identify the intermediates A, B, C, D and E in the following transformation. 5

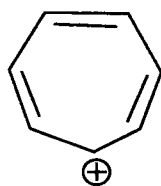


- ii) Phenyllithium removes two hydrogens as protons from the given compound. The dianion obtained in this process is stable. Provide an explanation for the stability of dianion. 4

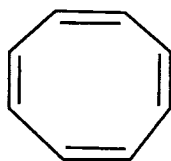




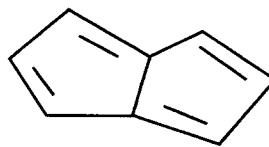
Q6. (a) iii) Identify whether the following structures are aromatic/antiaromatic/non aromatic. 6



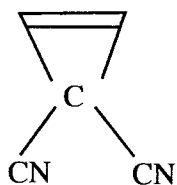
(I)



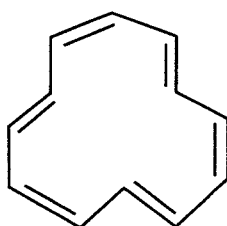
(II)



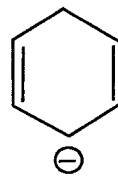
(III)



(IV)

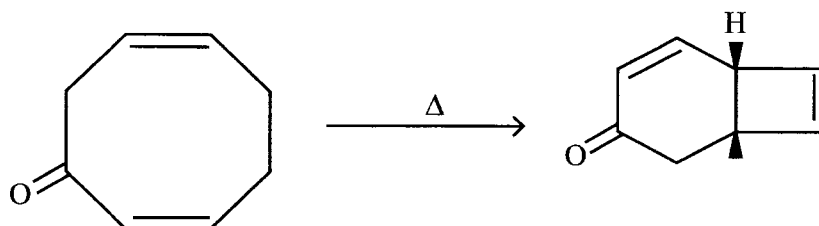


(V)

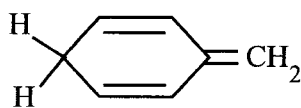


(VI)

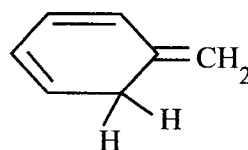
Q6. (b) i) Discuss the mechanism of the following electrocyclic reaction. 6



ii) Consider the following two isomeric compounds (A) and (B). 4



(A)



(B)

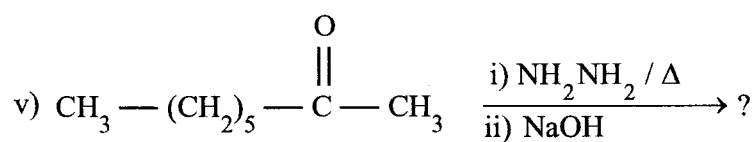
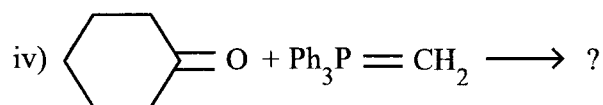
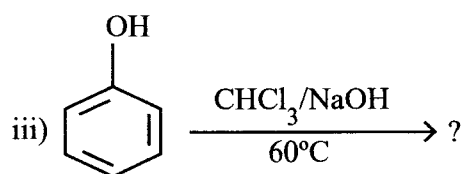
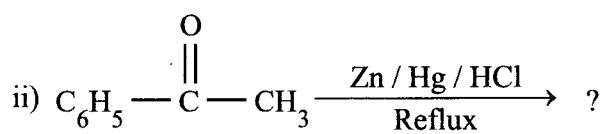
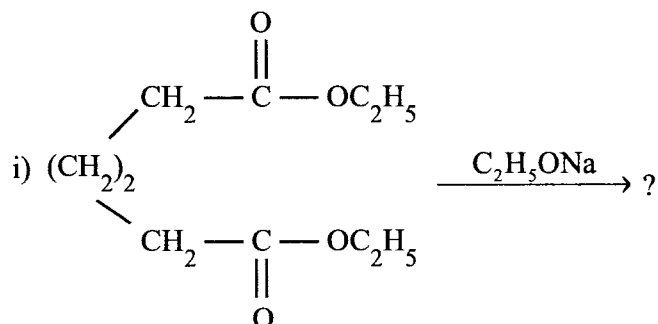
Compound (A) on heating readily converts into toluene but (B) does not. Explain this observation.

iii) What is Claisen rearrangement ? Give its mechanism. 5



Q6. (c) Name the following reactions and predict the product.

10



Q7. (a) Discuss the preparation, properties and uses of the polymers.

15

- 1) Polystyrene
- 2) Buna-s
- 3) Nylon
- 4) Polyethylene
- 5) Terylene

P.T.O.



Q7. (b) i) What is chemical shift ? Why TMS is used as reference standard ? 6

ii) Deduce the structure of the following organic molecules by using NMR spectral data. (3×3=9)

I) Molecular Formula : C_4H_9Br 3

NMR signals

a) 1.1 δ (d, 6H)

b) 1.95 δ (m, 1H)

c) 3.5 δ (d, 2H) 3

II) Molecular Formula : C_8H_8O

NMR signals

a) 7.27 δ (Singlet, 5H)

b) 2.8 δ (d, 2H)

c) 9.80 δ (t, 1H)

III) Molecular Formula : C_4H_8O 3

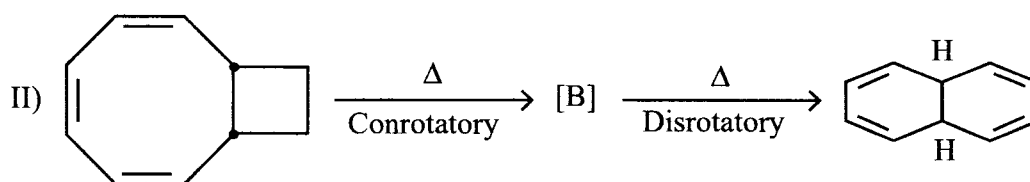
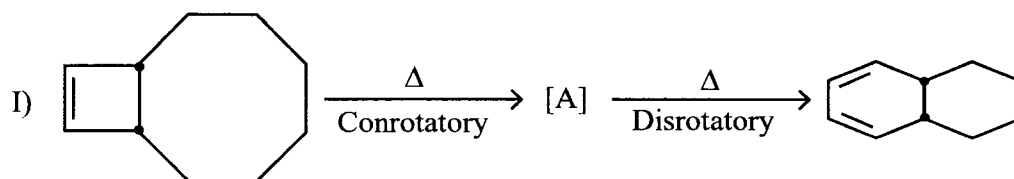
NMR signals

a) 2.48 δ (q, 2H)

b) 2.10 δ (s, 3H)

c) 1.07 δ (t, 3H)

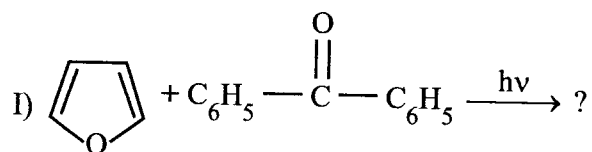
Q7. (c) i) Write the intermediates in following transformations. 2



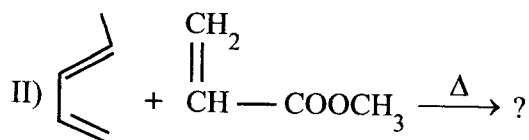


Q7. (c) ii) Complete the following reaction with mechanism.

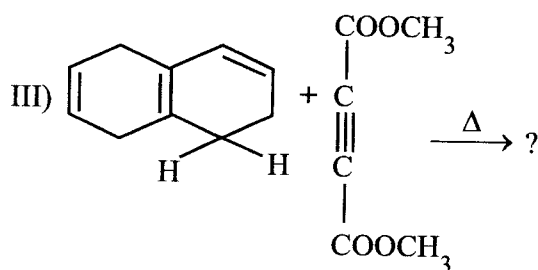
(2×4=8)



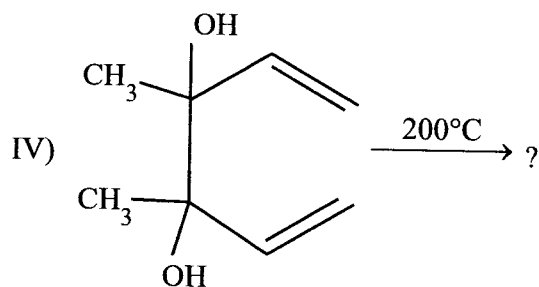
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2



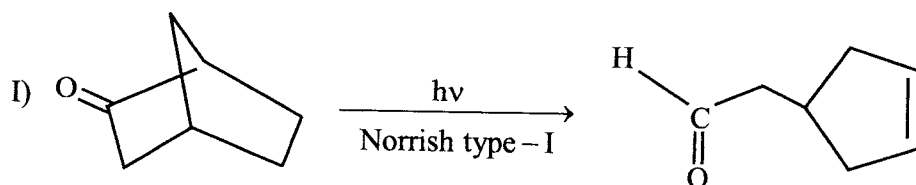
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2

Q8. (a) Give the mechanism of the following reactions.

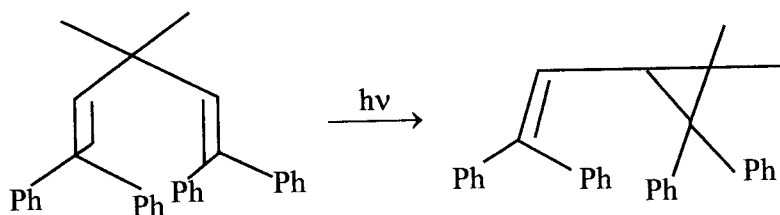
(3×5=15)



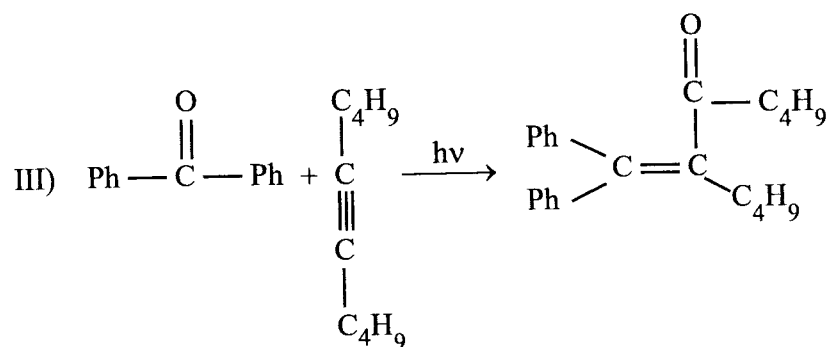
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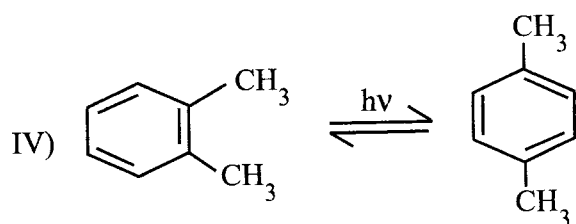
Q8. (a) II)



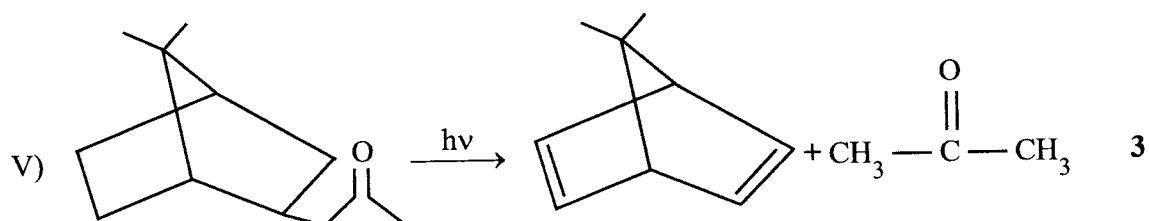
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3



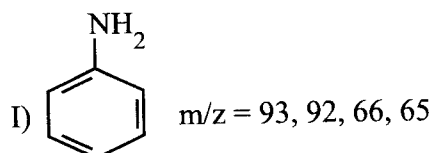
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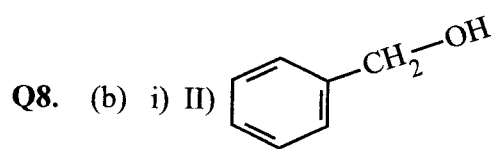
3

Q8. (b) i) Give the genesis of the following ions.

(3×4=12)

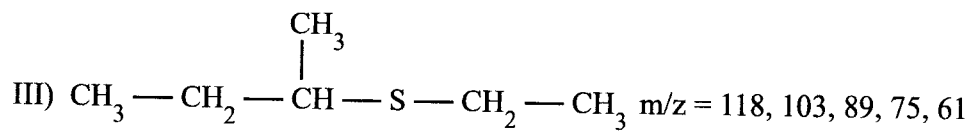


4



$m/z = 108, 106, 105, 77, 51$

4



4

ii) What is McLafferty rearrangement ? Explain with example.

3

Q8. (c) i) What is Perkin reaction ? Give its mechanism.

5

ii) What is Cannizzaro reaction ? Give its mechanism.

5

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